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United States Patent Application Publication

20250277574

Kind Code

A1

Publication Date

September 04, 2025

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ELECTRONIC APPARATUS AND METHOD OF MANUFACTURING ELECTRONIC APPARATUS

Abstract

display An electronic apparatus includes: an exterior body having, on a front surface, a display that shows at least one of a character, a symbol, or a figure; and a light-emitting unit that is provided inside the exterior body at a position facing the display and that is configured to emit light to an outside through the exterior body, in which a difference in transmittance of the light emitted from the light-emitting unit is generated between a region corresponding to the display and a region on a periphery in the exterior body.

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Appl. No.: 19/064607

Filed: February 26, 2025

Foreign Application Priority Data

JP 2024-031548

Mar. 01, 2024

Publication Classification

Int. Cl.: F21V9/08 (20180101); G03B27/00 (20060101)

U.S. Cl.:

CPC F21V9/08 (20130101); G03B27/00 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority under 35 USC 119 from Japanese Patent Application No. 2024-031548 filed on Mar. 1, 2024, the disclosure of which is incorporated by reference herein.

BACKGROUND

1. Technical Field

[0002] The present disclosed technology relates to an electronic apparatus and a method of manufacturing an electronic apparatus.

2. Related Art

[0003] JP2011-10101A discloses a portable electronic apparatus including: a housing; a light-emitting unit disposed inside the housing; and a light guide member having a light-receiving surface that faces the light-emitting unit and that receives light emitted from the light-emitting unit and a radiation surface that is exposed to the outside of the housing and radiates light, in which a surface of the light guide member other than the light-receiving surface and the radiation surface is subjected to a reflection treatment for reflecting light inside.

[0004] JP2011-250202A discloses a portable electronic apparatus comprising: a sheet substrate having light transmittance, in which an operation unit for a user operation is formed on a part of a region on a front surface of the sheet substrate, a substrate is provided on a back surface side of the sheet substrate, an information acquisition module for acquiring operation information that is changed in accordance with the user operation is formed in a region facing the operation unit on a surface of the substrate, a light-emitting element that illuminates the back surface of the sheet substrate is mounted in a region near an outer edge of the substrate and different from a region in which the information acquisition module is formed, a light shielding member extending to surround the substrate is installed on the back surface of the sheet substrate, a light shielding layer that blocks incidence of light on the sheet substrate is formed on the back surface of the sheet substrate, and the light shielding layer extends at least from a region facing the light-emitting element on the back surface of the sheet substrate to a region in which the light shielding member is installed toward an outer side of the region.

[0005] JP2012-104020A discloses an optical pad comprising: a pad main body; and a light source, in which the pad main body has a device placement surface that serves as an operation plane of a small-sized device comprising an imaging device, and a light and dark pattern is formed on the device placement surface by light emitted from the light source.

[0006] JP2001-330898A discloses an instant printer comprising: a loading chamber that stores a film pack in which unexposed instant films are stacked and in which a transport port for transporting the instant films one by one is formed; a first transport unit that transports the instant films transported from the transport port in a bent state; a second transport unit that transports the instant films transported by the first transport unit along an outer peripheral portion of the loading chamber; an exposure head that is disposed in the vicinity of the transport port and that irradiates the instant films transported by the first transport unit with light; and a spreading roller that spreads a developer by pressing the instant films transported by the second transport unit.

SUMMARY

[0007] In a case in which a display unit (for example, a logo or the like) is provided on an exterior body of the electronic apparatus, the visibility of the display unit may be decreased by light emitted from a light-emitting unit inside the exterior body.

[0008] Therefore, the present disclosed technology provides an electronic apparatus and a method of manufacturing an electronic apparatus, which are capable of improving the visibility of a display unit even in a case in which a light-emitting unit emits light.

[0009] A first aspect according to the present disclosed technology relates to an electronic apparatus comprising: an exterior body having, on a front surface, a display unit that shows at least one of a character, a symbol, or a figure; and a light-emitting unit that is provided inside the exterior body at a position facing the display unit and that is configured to emit light to an outside through the exterior body, in which a difference in transmittance of the light emitted from the light-emitting unit is generated between a region corresponding to the display unit and a region on a periphery in the exterior body.

[0010] A second aspect according to the present disclosed technology relates to the electronic apparatus according to the first aspect, in which a transmission suppression layer containing a material having lower light transmittance than the periphery is formed on an inner surface of the exterior body, to suppress transmission of the light in the region corresponding to the display unit.

[0011] A third aspect according to the present disclosed technology relates to the electronic apparatus according to the second aspect, in which the transmission suppression layer is formed by printing.

[0012] A fourth aspect according to the present disclosed technology relates to the electronic apparatus according to the second aspect, in which the transmission suppression layer is formed by attaching an attachment member.

[0013] A fifth aspect according to the present disclosed technology relates to the electronic apparatus according to the second aspect, in which the material having lower light transmittance is a coating material that absorbs light having a wavelength in a visible light range.

[0014] A sixth aspect according to the present disclosed technology relates to the electronic apparatus according to the first aspect, in which the display unit shows at least one of the character, the symbol, or the figure by using a difference in a three-dimensional shape from the periphery on the front surface.

[0015] A seventh aspect according to the present disclosed technology relates to the electronic apparatus according to the sixth aspect, in which the three-dimensional shape is a shape that protrudes to an outer side of the exterior body rather than the periphery.

[0016] An eighth aspect according to the present disclosed technology relates to the electronic apparatus according to the first aspect, in which a coating film is formed on an outer side of the exterior body, and a thickness of the coating film in a transmission portion through which the light from the light-emitting unit is transmitted is set to be less than a thickness of the coating film in another region.

[0017] A ninth aspect according to the present disclosed technology relates to the electronic apparatus according to the first aspect, in which a stripe pattern consisting of periodic recessed and protruding portions is formed on the front surface of the exterior body, and the difference in the three-dimensional shape is a difference in a height of the protruding portion of the stripe pattern.

[0018] A tenth aspect according to the present disclosed technology relates to the electronic apparatus according to the first aspect, in which the electronic apparatus has a printer function, and the light-emitting unit is provided at a position facing the display unit formed on a surface of the exterior body opposite to a side on which a film containing a photosensitive material is accommodated.

[0019] An eleventh aspect according to the present disclosed technology relates to a method of manufacturing an electronic apparatus, the method comprising: preparing an exterior body having, on a front surface, a display unit that shows at least one of a character, a symbol, or a figure, the exterior body emitting light from a light-emitting unit that is provided at a position facing the display unit to an outside through the exterior body; and performing processing of generating a difference in transmittance of the light emitted from the light-emitting unit between a region corresponding to the display unit and a region on a periphery in the exterior body.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] Exemplary embodiments of the technology of the disclosure will be described in detail based on the following figures, wherein:

[0021] FIG. **1** is a conceptual diagram illustrating an example of a configuration of an electronic apparatus according to an embodiment;

[0022] FIG. **2** is a plan view illustrating an example of the configuration of the electronic apparatus according to the embodiment;

[0023] FIG. **3** is a cross-sectional view illustrating an example of the configuration of the electronic apparatus according to the embodiment;

[0024] FIG. **4** is a conceptual diagram illustrating an example of the configuration of the electronic apparatus according to the embodiment;

[0025] FIG. **5** is a conceptual diagram illustrating an example of the configuration of the electronic apparatus according to the embodiment;

[0026] FIG. **6** is a conceptual diagram illustrating an example of the configuration of the electronic apparatus according to the embodiment;

[0027] FIG. **7** is a cross-sectional view illustrating an example of the configuration of the electronic apparatus according to the embodiment;

[0028] FIG. **8** is a side view illustrating an example of the configuration of the electronic apparatus according to the embodiment;

[0029] FIG. **9** is a flowchart illustrating an example of a processing process according to the embodiment;

[0030] FIG. **10** is a flowchart illustrating an example of a painting process according to the embodiment and

[0031] FIG. **11** is a conceptual diagram illustrating an example of the configuration of the electronic apparatus according to the embodiment.

DETAILED DESCRIPTION

[0032] An example of an embodiment of an electronic apparatus **10** according to the present disclosed technology will be described with reference to the accompanying drawings.

[0033] Hereinafter, for convenience of description, a width direction, a front-rear direction, and a height direction of the electronic apparatus **10** are indicated by three arrows X, Y, and Z. First, the height direction is indicated by the arrow Z, a direction indicated by the arrow Z is referred to as an up direction of the electronic apparatus **10**, and an opposite direction thereof is referred to as a down direction. The width direction is indicated by the arrow X orthogonal to the arrow Z, a direction indicated by the arrow X is referred to as a right direction of the electronic apparatus **10**, and an opposite direction thereof is referred to as a left direction. The front-rear direction is indicated by the arrow Y in a direction orthogonal to the arrow Z and the arrow X, a direction indicated by the arrow Y is referred to as a front direction of the electronic apparatus **10**, and an opposite direction thereof is referred to as a rear direction. In addition, in the following description, the expressions using the side, such as an upper side, a lower side, a left side, a right side, a front side, and a rear side, have the same meaning as the expressions using the direction.

[0034] As illustrated in FIG. **1** as an example, the electronic apparatus **10** has, for example, a printer function. The electronic apparatus **10** is, for example, a portable electronic apparatus. The electronic apparatus **10** is, for example, a mobile printer. Hereinafter, for convenience of description, the embodiment will be described with the mobile printer **11** as an example. The electronic apparatus **10** is an example of an “electronic apparatus” according to the present disclosed technology.

[0035] The mobile printer **11** can communicate with an external device. In the example illustrated

in FIG. 1, the mobile printer **11** can communicate with a smartphone **100**. Examples of a communication method include wireless communication (for example, Bluetooth (registered trademark)), but wired communication may be used. The mobile printer **11** receives image data IM transmitted from the smartphone **100** and prints a subject image shown by the received image data IM on a film F. The film F is, for example, a film containing a photosensitive material (so-called instant film). The film F is an example of a “film” according to the present disclosed technology. [0036] The mobile printer **11** comprises an exterior body **12**. In the example illustrated in FIG. 1, the exterior body **12** has a rectangular parallelepiped-shaped housing structure having a longitudinal direction in an X direction. The exterior body **12** accommodates, inside, electronic components and the like for implementing the printer function and other functions of the mobile printer **11**. In addition, for example, the printed film F is discharged from an outlet **14** provided on a left side surface of the exterior body **12**. The exterior body **12** is an example of an “exterior body” according to the present disclosed technology.

[0037] A display unit (display) **16** is formed on the exterior body **12**. The display unit **16** shows at least one of a character, a symbol, or a figure on a front surface of the exterior body **12**. In the example illustrated in FIG. 1, the display unit **16** is formed on an upper surface **12A** of the exterior body **12**. In the example illustrated in FIG. 1, the display unit **16** shows the characters “ABCD”. Examples of the content shown by the display unit **16** include a company name, a product name, a personal name, a catchphrase, a logo mark, and/or a pictogram. In addition, examples of the content shown by the display unit **16** may include a symbol (for example, an arrow indicating a film discharge direction) or a figure (for example, a mark of a power button) for describing an operation of the mobile printer **11**. The display unit **16** is an example of a “display unit” according to the present disclosed technology.

[0038] As illustrated in FIG. 2 as an example, a linear stripe pattern is formed on the exterior body **12** along the longitudinal direction (here, the X direction). Specifically, the stripe pattern consisting of periodic recessed and protruding portions is formed on the front surface of the exterior body **12**. Hereinafter, the stripe pattern is described with the upper surface **12A** on the front surface of the exterior body **12** as an example, but it goes without saying that the stripe pattern may be formed on other surfaces of the exterior body **12** in the same manner.

[0039] In a case of being viewed from a cross section having the longitudinal direction as a normal direction (in a cross-sectional view), protruding portions **18** are periodically formed on the upper surface **12A** of the exterior body **12**. In a cross-sectional view, the protruding portion **18** has a triangular shape, and a portion at a vertex is continuous along the longitudinal direction, whereby the linear stripe pattern is formed. As described above, the protruding portions **18** are periodically present, and thus the stripe pattern consisting of periodic recessed and protruding portions is formed.

[0040] Here, the display unit **16** shows at least one of the character, the symbol, or the figure due to a difference in a three-dimensional shape from a periphery on the front surface of the exterior body **12**. In the example illustrated in FIG. 2, the display unit **16** protrudes to an outer side (here, an upper side) of the exterior body **12** rather than the periphery. As a result, the difference in the three-dimensional shape from the periphery is generated. Specifically, in the above-described stripe pattern, a portion corresponding to the display unit **16** is the protruding portion **20** that is higher than the protruding portion **18** present on the periphery. That is, in the example illustrated in FIG. 2, the difference in the three-dimensional shape is a difference in a height of the protruding portion of the stripe pattern.

[0041] The difference in the three-dimensional shape between the display unit **16** and the other region causes a difference in a way of reflecting natural light (here, ambient light). In this way, the display unit **16** can be visually recognized due to the difference in the three-dimensional shape from the periphery (here, the characters “ABCD” appear to be floating).

[0042] As illustrated in FIG. 3 as an example, the exterior body **12** includes a first member **22** and a

second member **24** disposed adjacent to the first member **22**. The first member **22** is, for example, a cover member that constitutes the upper side of the exterior body **12**, and the second member **24** is, for example, a cover member that constitutes a lower side of the exterior body **12**. The mobile printer **11** comprises a light-emitting unit (light-emitting device) **26** inside the exterior body **12**. The light-emitting unit **26** is provided at a position facing the display unit **16** inside the exterior body **12**. The light-emitting unit **26** can emit emission light L to the outside through the exterior body **12**. The emission light L is, for example, light including a wavelength in a visible light range. The light-emitting unit **26** is an example of a “light-emitting unit” according to the present disclosed technology.

[0043] In the example illustrated in FIG. 3, the light-emitting unit **26** includes, for example, light-emitting diode (LED) light sources **26A** to **26C**. The LED light sources **26A** to **26C** are linearly arranged in an order from a right side inside the exterior body **12**. Here, the form example is described in which the three LED light sources **26A** to **26C** are disposed, but this is merely an example, and two or less or four or more LED light sources may be used. In addition, a disposition interval of the LED light sources, positions at which the LED light sources are disposed, and a disposition method (for example, disposing the LED light sources in a circular shape or disposing the LED light sources at positions of vertices of a quadrangle) can also be set as appropriate. In addition, each of the LED light sources **26A** to **26C** includes RGB light-emitting elements, and a color and a light amount of the emission light L can be changed by controlling a light emission aspect of each light-emitting element.

[0044] In addition, in the example illustrated in FIG. 3, the light-emitting unit **26** is accommodated inside the first member **22** that constitutes an upper portion of the exterior body **12**. The first member **22** comprises a contact portion **23** as a portion in contact with the second member **24**. Specifically, the first member **22** includes a bottom plate portion **22A** as a plate-shaped portion having a plate thickness direction in the height direction, and a side wall portion **22B** that is erected from a peripheral edge of the bottom plate portion **22A** along the height direction. Here, the side wall portion **22B** is erected toward the second member **24** in a state in which the first member **22** and the second member **24** are combined. Although the details will be described later, the side wall portion **22B** is in contact with the second member **24**, and the side wall portion **22B** also functions as the contact portion **23**.

[0045] The light-emitting unit **26** is provided at a position facing an inner surface of the bottom plate portion **22A**. The emission light L from the light-emitting unit **26** is transmitted through the bottom plate portion **22A** and is emitted to the outside of the exterior body **12**. Here, the transmission of the emission light L includes that the emission light L is scattered in the exterior body **12** and is emitted to the outside. Here, since the exterior body **12** is formed of a resin material, the emission light L is scattered in a case of passing through the bottom plate portion **22A**, and an emission range of the emission light L to the outside is widened.

[0046] In addition, a space **28** in which the film F is accommodatable is formed in the second member **24** constituting a lower portion of the exterior body **12**. The film F is accommodated inside the space **28** via a cover (not illustrated). Here, for convenience of description, one film F is illustrated, but it goes without saying that a plurality of films F may be accommodated in the space **28**. The space **28** can be open with respect to a lower surface of the exterior body **12**. In other words, the light-emitting unit **26** is provided at a position facing the display unit **16** formed on a surface (here, the upper surface) of the exterior body **12** opposite to a side on which the film F is accommodated. The second member **24** also has the same shape as the first member **22**.

[0047] As illustrated in FIG. 4 as an example, a transmission portion **36**, which is a region through which the emission light L is transmitted through the exterior body **12**, is determined by a distance between the light-emitting unit **26** and the inner surface of the exterior body **12** and a spread angle of the emission light L. Specifically, the transmission portion **36** is determined by the distance between the LED light sources **26A** to **26C** of the light-emitting unit **26** and an inner surface **22A1**

of the bottom plate portion **22A** and the spread angle of the emission light **L** from the LED light sources **26A** to **26C**. In other words, the first member **22** of the exterior body **12** has the transmission portion **36** that allows the transmission of the emission light **L** from the light-emitting unit **26**. The transmission portion **36** is an example of a “transmission portion” according to the present disclosed technology.

[0048] In the mobile printer **11**, the light-emitting unit **26** may emit light (for example, light up on or blink). Accordingly, it is possible to notify of an operation state of the mobile printer **11** or to improve a user experience (for example, to improve the fun that a user feels during use). Here, in a case in which the exterior body **12** of the mobile printer **11** is viewed in a plan view (here, in a case of being viewed from the upper surface), the light emission via the light-emitting unit **26** overlaps a position of the display unit **16** (that is, the light-emitting unit **26** faces the display unit **16**). In such a case, it is difficult to see the display unit **16** due to the light emission of the light-emitting unit **26**. That is, in a case in which the light-emitting unit **26** emits light, the emission light **L** transmitted through the exterior body **12** is easily visually recognized, and reflected light (for example, reflected light caused by the difference in three-dimensional shape from the periphery) in the display unit **16** is not easily visually recognized. As a result, the visibility of the display unit **16** is decreased.

[0049] Therefore, as illustrated in FIG. 5 as an example, in the present embodiment, in the exterior body **12**, a difference in transmittance of the emission light **L** emitted from the light-emitting unit **26** is generated between a region corresponding to the display unit **16** and a region on the periphery. Specifically, a transmission suppression layer **30** containing a material having lower light transmittance than the periphery is formed on the inner surface of the exterior body **12**. More specifically, the transmission suppression layer **30** is formed on the inner surface **22A1** of the bottom plate portion **22A** in the first member **22** of the exterior body **12**. The transmission suppression layer **30** is formed in the region corresponding to the display unit **16**. The region corresponding to the display unit **16** is, for example, a back surface of the display unit **16**, and is a range having the same size as the display unit **16**. In the example illustrated in FIG. 5, transmission suppression layers **30A**, **30B**, **30C**, and **30D** having shapes in which the characters “ABCD” shown by the display unit **16** are inverted are formed. The region corresponding to the display unit **16** need not be a range having the same size as the display unit **16**, and the size may be changed as appropriate. The transmission suppression layers **30**, **30A**, **30B**, **30C**, and **30D** are examples of a “transmission suppression layer” according to the present disclosed technology.

[0050] The transmission suppression layers **30A**, **30B**, **30C**, and **30D** are formed on the inner surface **22A1** of the first member **22** by printing. Specifically, the transmission suppression layers **30A**, **30B**, **30C**, and **30D** are formed by printing using a coating material containing a material that absorbs the wavelength in the visible light range. For example, the transmission suppression layers **30A**, **30B**, **30C**, and **30D** are printed using a black coating material. A part of the emission light **L** is absorbed or reflected due to the interaction with the transmission suppression layer **30**. In this way, the transmission of the emission light **L** in the region corresponding to the display unit **16** is suppressed.

[0051] As described above, the light-emitting unit **26** is provided at a position facing the display unit **16**. In addition, inside the exterior body **12**, the transmission suppression layer **30** is formed in the region corresponding to the display unit **16**. Therefore, the transmission of a part of the emission light **L** from the light-emitting unit **26** is suppressed by the transmission suppression layer **30**. Here, a part of the emission light **L** is absorbed by the black coating material included in the transmission suppression layer **30**. In addition, in the example illustrated in FIG. 5, in a case in which the first member **22** is viewed from the inner surface side (here, in a case of being viewed from below), the transmission suppression layers **30A**, **30B**, **30C**, and **30D** have the shapes in which the characters “ABCD” are inverted. Therefore, in the region having the shapes in which the characters of “ABCD” are inverted, the transmission of the emission light **L** through the exterior

body **12** is suppressed.

[0052] As illustrated in FIG. **6** as an example, in the mobile printer **11**, in a case in which the light-emitting unit **26** emits light, as described above, the transmission of the emission light **L** is suppressed by the transmission suppression layer **30** provided in the region corresponding to the display unit **16**. As a result, the emission light **L** is less likely to be transmitted through the display unit **16**, and thus the display unit **16** is easily visually recognized. In the example illustrated in FIG. **6**, the characters “ABCD” of the display unit **16** are easily visually recognized by the transmission suppression layer **30** having the shapes in which the characters “ABCD” are inverted. In addition, in a range illuminated with the emission light **L**, the content (here, the characters “ABCD”) indicated by the transmission suppression layer **30** floats as a shadow, and the designability is improved.

[0053] A portion of the exterior body **12** facing the light-emitting unit **26** is required to contain a material having relatively high transmittance in order to transmit the emission light **L** from the light-emitting unit **26**. On the other hand, a portion opposite to the portion facing the light-emitting unit **26** is required to contain a material having relatively low transmittance. This configuration is to suppress the entry of unnecessary external light into the inside of the exterior body **12**.

[0054] In a case in which the exterior body **12** is formed by combining two types of members having different optical characteristics (for example, light transmittance), a difference in the optical characteristics makes it noticeable that the two types of members are separate components in a portion in which the two types of members are in contact with each other. As a result, the exterior body **12** may lack a sense of unity as a whole and may impair the exterior characteristics.

[0055] Therefore, as illustrated in FIG. **7** as an example, a thickness of a coating film **32** is adjusted in the first member **22** of the exterior body **12**. The first member **22** includes the coating film **32** and a substrate **34**. The coating film **32** is formed on a front surface of the substrate **34**. The substrate **34** is formed of a translucent material. Here, the translucent material is a material that allows the transmission of the light while scattering the light. The substrate **34** is, for example, a milky white resin. In addition, the coating film **32** is a film formed by fixing the coating material to the front surface of the substrate **34**, and the type of the coating material is set as appropriate in accordance with a color variation of the exterior body **12**. As described above, the emission light **L** from the light-emitting unit **26** is emitted to the outside through the first member **22**. Since the first member **22** is formed of the translucent material, the emission light **L** is easily emitted to the outside of the exterior body **12**, and the emission light **L** is easily visually recognized. The coating film **32** is an example of a “coating film” according to the present disclosed technology.

[0056] The coating film **32** has a thickness t_1 in the side wall portion **22B** of the first member **22**. On the other hand, the coating film **32** has a thickness t_2 in the bottom plate portion **22A** of the first member **22**. In addition, the thickness t_1 of the coating film **32** in the side wall portion **22B** is greater than the thickness t_2 of the coating film **32** in the bottom plate portion **22A**. In other words, in the transmission portion **36** (see FIG. **4**) through which the emission light **L** is transmitted in the bottom plate portion **22A**, the thickness t_2 of the coating film **32** is less than the thickness t_1 of the coating film **32** in the side wall portion **22B**.

[0057] Specifically, the thickness t_1 of the coating film **32** in the side wall portion **22B** is 1.5 times or more the thickness t_2 of the coating film **32** in the bottom plate portion **22A**. More specifically, the thickness t_1 is, for example, about 20 μm , and the thickness t_2 is, for example, about 10 to 12 μm .

[0058] The second member **24** includes a bottom plate portion **24A** and a side wall portion **24B** that is erected from a peripheral edge of the bottom plate portion **24A** along the height direction. The second member **24** is formed of a non-transparent material. Here, the non-transparent material is a material that cuts light. Specifically, the non-transparent material is, for example, a material in which titanium oxide particles are dispersed in a resin as a base material. That is, the second member **24** is formed of, for example, a resin containing titanium oxide. As described above, the

space **28** in which the film **F** is accommodated is formed in the second member **24**. Since the second member **24** is formed of the non-transparent material, the entry of the external light into the inside of the second member **24** is suppressed.

[0059] The first member **22** and the second member **24** are joined in a state of being adjacent to each other. Further, the first member **22** is in contact with the second member **24** via the contact portion **23**. In the example illustrated in FIG. 7, the side wall portion **22B** of the first member **22** and the side wall portion **24B** of the second member **24** are in contact with each other. That is, in the first member **22**, the side wall portion **22B** is the contact portion **23**. Specifically, the lower surface of the side wall portion **22B** of the first member **22** and the upper surface of the side wall portion **24B** of the second member **24** are in contact with each other. Then, as described above, the thickness t_1 of the coating film **32** in the side wall portion **22B** is greater than the thickness t_2 of the coating film **32** in the bottom plate portion **22A**. That is, the thickness t_1 of the coating film **32** in the contact portion **23** is greater than the thickness t_2 of the coating film **32** in the transmission portion **36**.

[0060] The first member **22** and the second member **24** are joined to each other in a state in which the side wall portion **22B** and the side wall portion **24B** are in contact with each other. The method of joining the first member **22** and the second member **24** is not particularly limited, and may be, for example, mechanical fastening in which a fixing unit for fitting a claw for fixing into a hole and screwing are combined, or may be joined by an adhesive, ultrasonic welding, or the like.

[0061] As an example, a case is considered in which the thickness t_1 of the coating film **32** in the side wall portion **22B** of the first member **22** (hereinafter, simply referred to as a coating film thickness t_1) is less than the thickness t_2 of the coating film **32** in the bottom plate portion **22A** of the first member **22** (hereinafter, simply referred to as a coating film thickness t_2), as illustrated on the left side of FIG. 8. In such a case, in a portion (here, the side wall portion **22B** and the side wall portion **24B**) in which the first member **22** and the second member **24** are in contact with each other, the difference in the light transmittance makes it noticeable that the first member **22** and the second member **24** are separate components.

[0062] On the other hand, as illustrated on the right side of FIG. 8, a case is considered in which the coating film thickness t_1 in the side wall portion **22B** of the first member **22** is greater than the coating film thickness t_2 in the bottom plate portion **22A** of the first member **22**. In such a case, the light transmittance of the side wall portion **22B** of the first member **22** is close to the light transmittance of the side wall portion **24B** of the second member **24**, so that a difference between the first member **22** and the second member **24** is less noticeable. As a result, a sense of unity of the entire exterior body **12** as a whole is increased, and the exterior characteristics are improved.

[0063] Hereinafter, a processing process and a painting process according to the present embodiment will be described with reference to FIGS. 9 and 10. The processing process and the painting process are parts of a manufacturing process of the mobile printer **11**. The flow of the processing process illustrated in FIG. 9 is an example of a “method of manufacturing an electronic apparatus” according to the present disclosed technology.

[0064] As illustrated in FIG. 9 as an example, in the processing process, first, in step ST12, the exterior body **12** including the display unit **16** is prepared. After the process of the step in step ST12 is performed, the processing process proceeds to step ST14.

[0065] In step ST14, processing of causing the difference in the transmittance of the light L (that is, the emission light L) emitted from the light-emitting unit **26** is performed in the region corresponding to the display unit **16** of the exterior body **12**. Specifically, the transmission suppression layer **30** is formed by printing in a region corresponding to the display unit **16** of the exterior body **12** on the inner surface of the exterior body **12**. After the process of the step in step ST14 is performed, the processing process ends.

[0066] As illustrated in FIG. 10 as an example, in the painting process, first, in step ST22, the first member **22** of the exterior body **12** is prepared. After the process of the step in step ST22 is

performed, the painting process proceeds to step ST24.

[0067] In step ST24, in the first member 22, the coating film 32 of the side wall portion 22B is provided to be thicker than the coating film 32 of the bottom plate portion 22A. After the process of the step in step ST24 is performed, the painting process ends.

[0068] As described so far, in the electronic apparatus 10 according to the present embodiment, the exterior body 12 includes the display unit 16. The display unit 16 shows the characters or the like on the front surface of the exterior body 12. Further, the light-emitting unit 26 is provided inside the exterior body 12 at a position facing the display unit 16. The light-emitting unit 26 emits the light L to the outside through the exterior body 12. Then, a difference in the transmittance of the light L emitted from the light-emitting unit 26 is generated between a region corresponding to the display unit 16 and a region of the periphery in the exterior body 12. As a result, even in a case in which the region corresponding to the display unit 16 is identified from other portions due to a difference in a degree of visual recognition and the light-emitting unit 26 provided inside the exterior body 12 is lighted up, it is easy to recognize the display unit 16.

[0069] In addition, in the electronic apparatus 10 according to the present embodiment, the transmission suppression layer 30 containing a material having lower light transmittance than the periphery is formed on the inner surface of the exterior body 12, so that the transmission of the emission light L in the region corresponding to the display unit 16 is suppressed. As a result, for example, the influence on the appearance (for example, the influence on the appearance of the display unit 16) is reduced as compared to a case in which the transmission suppression layer 30 is formed on an outer surface of the exterior body 12.

[0070] In the electronic apparatus 10 according to the present embodiment, the transmission suppression layer 30 is formed by printing. As a result, for example, the productivity of the electronic apparatus 10 can be improved as compared to a case in which the transmission suppression layer 30 is formed by attaching a seal.

[0071] In addition, in the electronic apparatus 10 according to the present embodiment, the material having lower light transmittance is a coating material that absorbs light having the wavelength in the visible light range. As a result, the transmission suppression layer 30 absorbs light having a wavelength in a visible light range emitted from the light-emitting unit 26, so that the visibility of the display unit 16 can be further improved.

[0072] In addition, in the electronic apparatus 10 according to the present embodiment, the display unit 16 shows at least one of the character, the symbol, or the figure by the difference in the three-dimensional shape from the periphery on the front surface of the exterior body 12. As a result, since the character or the like is shown on the exterior body 12 by the difference in the three-dimensional shape, the designability of the appearance of the exterior body 12 is improved.

[0073] In addition, in the electronic apparatus 10 according to the present embodiment, the three-dimensional shape has a shape that protrudes to the outer side of the exterior body 12 rather than the periphery. As a result, for example, as compared to a case in which the three-dimensional shape is recessed to the inner side of the exterior body 12, it is possible to secure a distance between the light-emitting unit 26 and the display unit 16 and to contribute to the reduction in size of the exterior body 12 while widening the emission range of the emission light L.

[0074] In addition, in the electronic apparatus 10 according to the present embodiment, in the exterior body 12, the coating film 32 is formed on the outer side, and the thickness t1 of the coating film 32 in the side wall portion 22B is greater than the thickness t2 of the coating film 32 in the bottom plate portion 22A. In other words, the thickness t2 of the coating film 32 in the transmission portion 36 through which the light from the light-emitting unit 26 is transmitted is set to be less than the thickness t1 of the coating film 32 in another region. Accordingly, since the emission light L from the light-emitting unit 26 is likely to be transmitted, even in a case in which the thickness of the coating film 32 is partially reduced, in the present configuration, it is easy to improve the visibility of the display unit 16. That is, since the thickness of the coating film 32 is small, the

emission light L is easily recognized, but a difference in the visibility is generated between the region corresponding to the display unit **16** and the other region, so that the visibility of the display unit **16** can be improved.

[0075] In addition, in the electronic apparatus **10** according to the present embodiment, the stripe pattern consisting of the periodic recessed and protruding portions is formed on the front surface of the exterior body **12**, and the difference in the three-dimensional shape in the display unit **16** is the difference in the height of the protruding portion of the stripe pattern. In a case in which the stripe pattern is formed on the front surface of the exterior body **12** in order to enhance the designability, the display unit **16** is formed by the difference in the height of the three-dimensional shape. In such a case, the emission light L is likely to be transmitted from the valleys of the recessed and protruding portions of the stripe pattern, but, in the present configuration, the display unit **16** is likely to be noticeable.

[0076] In addition, in the electronic apparatus **10** according to the present embodiment, the electronic apparatus **10** has the printer function. Further, in the exterior body **12**, the film F containing the photosensitive coating material is accommodated in the second member **24** on the lower side. The light-emitting unit **26** is provided at a position facing the display unit **16** formed on the upper surface **12A** of the exterior body **12**. As a result, it is possible to suppress the influence of the light emission of the light-emitting unit **26** on the film F accommodated inside the exterior body **12**.

Modification Example

[0077] In the above-described embodiment, the form example is described in which the transmission suppression layer **30** is formed by printing, but the present disclosed technology is not limited to this. In the present modification example, the transmission suppression layer **30** is formed by attaching the seal member **40**.

[0078] As illustrated in FIG. **11** as an example, in the present modification example, in the exterior body **12**, the transmission suppression layer **30** containing a material having lower light transmittance than the periphery is formed on the inner surface of the exterior body **12**.

Specifically, the seal member **40** is attached to the inner surface **22A1** of the bottom plate portion **22A** in the first member **22** of the exterior body **12**. The seal member **40** has, for example, a laminated structure and includes a pressure-sensitive adhesive layer, the transmission suppression layer **30**, and a coating layer. The seal member **40** is attached to the exterior body **12** via the pressure-sensitive adhesive layer. The transmission suppression layer **30** corresponding to the content shown by the display unit **16** is formed on the pressure-sensitive adhesive layer. The transmission suppression layer **30** contains the material that absorbs the wavelength in the visible light range. Further, the transmission suppression layer **30** is protected by the coating layer. The seal member **40** is attached to form the transmission suppression layer **30** in the region corresponding to the display unit **16**. The seal member **40** is an example of an “attachment member” according to the present disclosed technology.

[0079] In the example illustrated in FIG. **11**, the transmission suppression layers **30A**, **30B**, **30C**, and **30D** having shapes in which the characters “ABCD” shown by the display unit **16** are inverted are formed. The seal member **40** is attached to the inner surface **22A1** of the first member **22** of the exterior body **12**. In this way, the transmission of the emission light L in the region corresponding to the display unit **16** is suppressed. As a result, the difference in the transmittance of the emission light L emitted from the light-emitting unit **26** is generated between the region corresponding to the display unit **16** and the region on the periphery.

[0080] As described so far, in the electronic apparatus **10** according to the present modification example, the transmission suppression layer **30** is formed by attaching the seal member **40**. As a result, in a case in which the seal member **40** is attached, the transmission suppression layer **30** can be formed, which can contribute to reduction in manufacturing cost of the electronic apparatus **10**.

[0081] It should be noted that, in the above-described embodiment, the form example is described

in which the display unit **16** is provided on the upper surface **12A** of the exterior body **12**, but the present disclosed technology is not limited to this. The display unit **16** may be provided on other surfaces of the exterior body **12**. In addition, a plurality of display units **16** may be provided on the front surface of the exterior body **12**.

[0082] In addition, in the above-described embodiment, the form example is described in which the three-dimensional shape of the display unit **16** protrudes rather than the periphery, but the present disclosed technology is not limited to this. For example, the three-dimensional shape of the display unit **16** may be a shape that is recessed rather than the periphery, and the periphery may be roughened and the display unit **16** may be a smooth surface.

[0083] In addition, in the above-described embodiment, the form example is described in which the transmission suppression layer **30** is formed by printing, but the present disclosed technology is not limited to this. For example, the transmission suppression layer **30** may be formed by being partially molded with a resin that is less likely to allow the transmission of the light, or may be formed by attaching a light shielding cover of a separate component to the inside of the exterior body **12**.

[0084] In addition, in the above-described embodiment, the form example is described in which the transmission of the emission light **L** is suppressed in the region corresponding to the display unit **16**, but the present disclosed technology is not limited to this. For example, an aspect may be adopted in which the transmission of the emission light **L** is promoted in the region corresponding to the display unit **16**. Specifically, the transmission suppression layer **30** may be formed in a portion other than the region corresponding to the display unit **16**, to promote the transmission of light in the region corresponding to the display unit **16**. In this way, the difference in the light transmittance is generated between the region corresponding to the display unit **16** and the periphery.

[0085] In addition, in the above-described embodiment, the form example is described in which the stripe pattern having the periodic recessed and protruding portions is formed on the front surface of the exterior body **12**, but the present disclosed technology is not limited to this. The front surface of the exterior body **12** may have a structure that enhances the designability other than the stripe pattern, or may be a smooth surface.

[0086] In addition, in the above-described embodiment, the form example is described in which the first member **22** and the second member **24** have a symmetrical shape in a cross-sectional view, but the present disclosed technology is not limited to this. For example, the first member **22** may include the side wall portion **22B**, and the second member **24** may be a plate-like member. In such a case, an end surface of the side wall portion **22B** is in contact with the peripheral edge of the second member **24**.

[0087] In addition, in the above-described embodiment, the form example is described in which the exterior body **12** has a rectangular parallelepiped shape, but the present disclosed technology is not limited to this. For example, the exterior body **12** may have a square shape, a spherical shape, or a cylindrical shape.

[0088] In addition, in the above-described embodiment, the form example is described in which the electronic apparatus **10** is the mobile printer **11**, but the present disclosed technology is not limited to this. The electronic apparatus **10** may be, for example, a portable electronic apparatus, such as an instant camera, a digital camera, a smartphone, a tablet terminal, a laptop computer, or a mobile battery. In addition, the electronic apparatus **10** may be a stationary electronic apparatus, such as a desktop computer, a printer, a television, or a display.

[0089] In addition, in the above-described embodiment, the form example is described in which the film **F** is a so-called instant film containing the photosensitive material, but the present disclosed technology is not limited to this. For example, printing paper containing a heat sensitive material may be used instead of the film **F**.

[0090] In addition, in the above-described embodiment, the form example is described in which the light-emitting unit **26** includes the LED light source, but the present disclosed technology is not

limited to this. For example, the light-emitting unit **26** may include an organic electroluminescence (EL) element.

[0091] In addition, in the above-described embodiment, the form example is described in which the coating film **32** is thick as a whole in the side wall portion **22B**, but the present disclosed technology is not limited to this. The coating film **32** on half of the side wall portion **22B** on the second member **24** side may be thick.

[0092] In addition, in the above-described embodiment, the form example is described in which the coating film **32** is thin as a whole in the bottom plate portion **22A**, but the present disclosed technology is not limited to this. The coating film **32** may be thin in an elliptical region including the transmission portion **36**, or may be thin only in a region corresponding to the transmission portion **36**.

[0093] In addition, in the above-described embodiment, the form example is described in which the substrate **34** of the first member **22** is formed of the translucent material, but the present disclosed technology is not limited to this. For example, the substrate **34** may be the transparent material. However, as described above, it is preferable that the substrate **34** is formed of the translucent material from the viewpoint that the appearance adjustment by the coating film **32** is facilitated.

[0094] In addition, in the above-described embodiment, the form example is described in which the second member **24** is formed of the resin of the non-transparent material, but the present disclosed technology is not limited to this. For example, the second member **24** may also have a configuration in which a coating film (for example, a coating film containing titanium oxide) is formed on a substrate of the translucent material. However, as described above, it is desirable that the second member **24** is formed of the non-transparent material from the viewpoint that the entry of the external light can be suppressed.

[0095] The above-described contents and the above-illustrated contents are detailed descriptions of portions related to the present disclosed technology, and are merely examples of the present disclosed technology. For example, the description of the configuration, the function, the operation, and the effect are the description of examples of the configuration, the function, the operation, and the effect of the portion according to the present disclosed technology. Therefore, it goes without saying that unnecessary portions may be deleted, new elements may be added, or replacements may be made with respect to the above-described contents and the above-illustrated contents within a range that does not deviate from the scope of the present disclosed technology. Further, the description of, for example, common technical knowledge that does not need to be particularly described to enable the implementation of the present disclosed technology is omitted in the above-described contents and the above-illustrated contents in order to avoid confusion and to facilitate the understanding of the portions related to the present disclosed technology.

[0096] In the present specification, “A and/or B” is synonymous with “at least one of A or B”. That is, “A and/or B” may mean only A, only B, or a combination of A and B. In the present specification, the same concept as “A and/or B” also applies to a case in which three or more matters are expressed by association with “and/or”.

[0097] All of the documents, the patent applications, and the technical standards described in the present specification are incorporated into the present specification by reference to the same extent as in a case in which the individual documents, patent applications, and technical standards are specifically and individually noted to be described by reference.

[0098] In regard to the embodiment described above, the following supplementary notes will be further disclosed.

Supplementary Note 1

[0099] An electronic apparatus comprising: an exterior body having, on a front surface, a display unit that shows at least one of a character, a symbol, or a figure; and a light-emitting unit that is provided inside the exterior body at a position facing the display unit and that is configured to emit light to an outside through the exterior body, in which a difference in transmittance of the light

mitted from the light-emitting unit is generated between a region corresponding to the display unit and a region on a periphery in the exterior body.

Supplementary Note 2

[0100] The electronic apparatus according to supplementary note 1, in which a transmission suppression layer containing a material having lower light transmittance than the periphery is formed on an inner surface of the exterior body, to suppress transmission of the light in the region corresponding to the display unit.

Supplementary Note 3

[0101] The electronic apparatus according to supplementary note 2, in which the transmission suppression layer is formed by printing.

Supplementary Note 4

[0102] The electronic apparatus according to supplementary note 2, in which the transmission suppression layer is formed by attaching an attachment member.

Supplementary Note 5

[0103] The electronic apparatus according to any one of supplementary notes 2 to 4, in which the material having lower light transmittance is a coating material that absorbs light having a wavelength in a visible light range.

Supplementary Note 6

[0104] The electronic apparatus according to any one of supplementary notes 1 to 5, in which the display unit shows at least one of the character, the symbol, or the figure by using a difference in a three-dimensional shape from the periphery on the front surface.

Supplementary Note 7

[0105] The electronic apparatus according to supplementary note 6, in which the three-dimensional shape is a shape that protrudes to an outer side of the exterior body rather than the periphery.

Supplementary Note 8

[0106] The electronic apparatus according to any one of supplementary notes 1 to 7, in which a coating film is formed on an outer side of the exterior body, and a thickness of the coating film in a transmission portion through which the light from the light-emitting unit is transmitted is set to be less than a thickness of the coating film in another region.

Supplementary Note 9

[0107] The electronic apparatus according to supplementary note 6, in which a stripe pattern consisting of periodic recessed and protruding portions is formed on the front surface of the exterior body, and the difference in the three-dimensional shape is a difference in a height of the protruding portion of the stripe pattern.

Supplementary Note 10

[0108] The electronic apparatus according to any one of supplementary notes 1 to 9, in which the electronic apparatus has a printer function, and the light-emitting unit is provided at a position facing the display unit formed on a surface of the exterior body opposite to a side on which a film containing a photosensitive material is accommodated.

Claims

1. An electronic apparatus comprising: an exterior body having, on a front surface, a display that shows at least one of a character, a symbol, or a figure; and a light-emitting device that is provided inside the exterior body at a position facing the display and that is configured to emit light to an outside through the exterior body, wherein a difference in transmittance of the light emitted from the light-emitting device is generated between a region corresponding to the display and a region on a periphery in the exterior body.

2. The electronic apparatus according to claim 1, wherein a transmission suppression layer containing a material having lower light transmittance than the periphery is formed on an inner

surface of the exterior body, to suppress transmission of the light in the region corresponding to the display.

3. The electronic apparatus according to claim 2, wherein the transmission suppression layer is formed by printing.

4. The electronic apparatus according to claim 2, wherein the transmission suppression layer is formed by attaching an attachment member.

5. The electronic apparatus according to claim 2, wherein the material having lower light transmittance is a coating material that absorbs light having a wavelength in a visible light range.

6. The electronic apparatus according to claim 1, wherein the display shows at least one of the character, the symbol, or the figure by using a difference in a three-dimensional shape from the periphery on the front surface.

7. The electronic apparatus according to claim 6, wherein the three-dimensional shape is a shape that protrudes to an outer side of the exterior body rather than the periphery.

8. The electronic apparatus according to claim 1, wherein a coating film is formed on an outer side of the exterior body, and a thickness of the coating film in a transmission portion through which the light from the light-emitting device is transmitted is set to be less than a thickness of the coating film in another region.

9. The electronic apparatus according to claim 6, wherein a stripe pattern consisting of periodic recessed and protruding portions is formed on the front surface of the exterior body, and the difference in the three-dimensional shape is a difference in a height of the protruding portion of the stripe pattern.

10. The electronic apparatus according to claim 1, wherein the electronic apparatus has a printer function, and the light-emitting device is provided at a position facing the display formed on a surface of the exterior body opposite to a side on which a film containing a photosensitive material is accommodated.

11. A method of manufacturing an electronic apparatus, the method comprising: preparing an exterior body having, on a front surface, a display that shows at least one of a character, a symbol, or a figure, the exterior body emitting light from a light-emitting device that is provided at a position facing the display to an outside through the exterior body; and performing processing of generating a difference in transmittance of the light emitted from the light-emitting device between a region corresponding to the display and a region on a periphery in the exterior body.
